

Image processing of X-ray radiographs of planar targets at OMEGA

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1 Loading raw data and obtaining strips

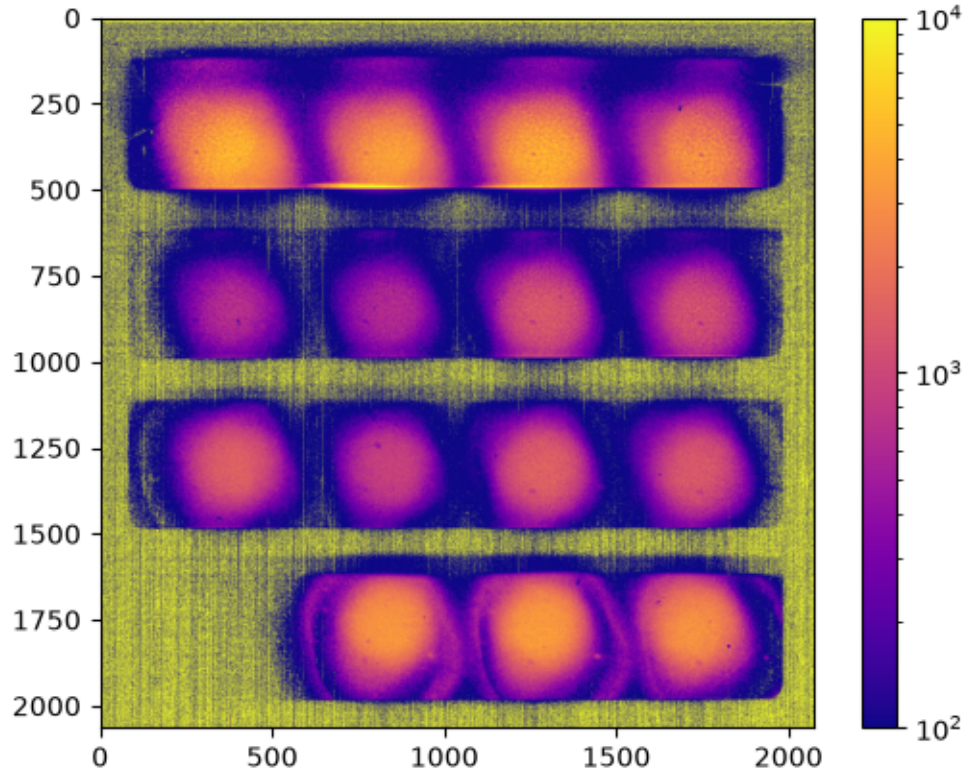


Figure 1: Full set of radiographic images for shot 117321. Each strip represents an image in time, with the time step increasing vertically

All data is processed in python. The data is first loaded in from a .hdf file, containing all data for each level of SSD. The data shown in Fig.1 is the lowest level of SSD, shot 117321. The yellow noise observed is a consequence of digitisation and can be ignored. When we say ignore we mean ‘set to zero’. The first step of the method is to obtain individual strips in order to complete analysis on individual shots. For this report, the strips will be numbered 1-4, from the bottom strip to the top strip.

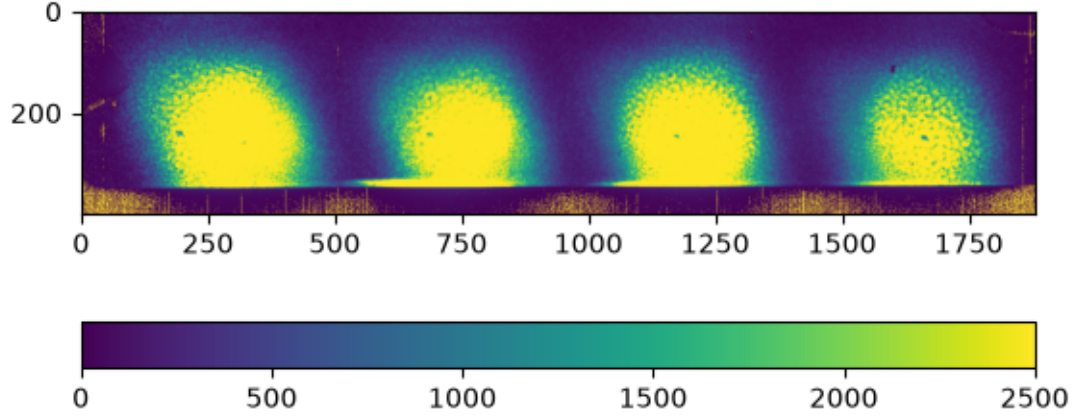


Figure 2: Strip 4 of shot 117321. Colour reflects pixel intensity in arbitrary units.

At later time steps there will be the most developed features, as instabilities have time to develop. For this methodology, we will use strip 4 from Fig.1 as an example, as shown in Fig.2. We can already observe noticeable features.

In order to do our data analysis, all images are stored as numpy arrays, with every entry being the intensity at that pixel. The pixel size in the image corresponds to $18\mu m$, however there is a magnification of $\times 6$, hence the physical size is $3\mu m$. Each strip has dimensions 400×1880 . For consistency, individual shots are cropped to 400×400 , as show in below in Fig.3. Shots will be numbered 1-4 from left to right.

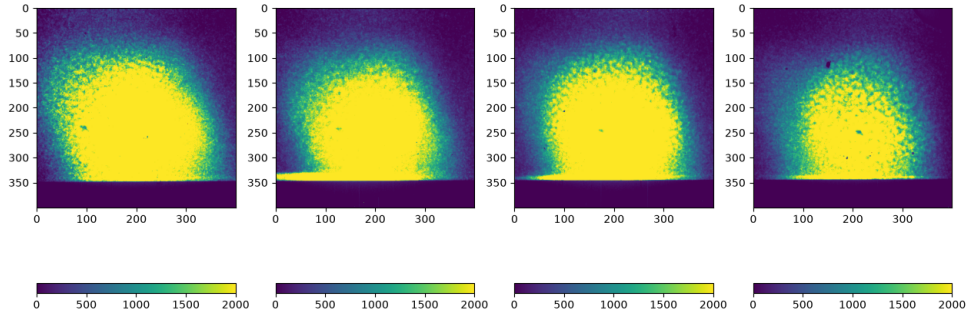


Figure 3: Individual shots from strip 4 of shot 117321.

2 Preliminary Noise Reduction

The shots shown in Fig.3 cannot be used for data analysis. They have far too much noise from several factors which makes it unsuitable for physical analysis. In order to remove the noise, we must remove the effects of following factors in the order of discussion.

2.1 Background Removal

Luca Antonelli wrote a very useful report regarding both background removal and back-lighter filtering. I have replicated his methodology for a 1D planar target.

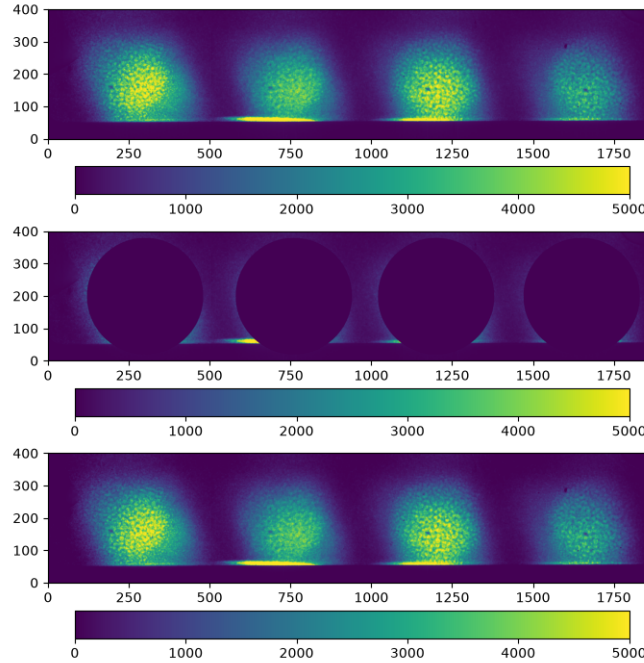


Figure 4: Individual shots from strip 4 of shot 117321.

To remove the background noise picked up by the CCD camera, we first

block out the shots on the strip by producing circular masks which overlap the numpy array, setting those regions to zero. We then choose a region of roughly uniform noise. We average this value, and then fill a 400×1880 numpy array with this value. We then take this array away from our strip image to remove the background noise. Fig.4 outlines this procedure, and shows the resultant ‘clean’ strip.

2.2 Back lighter profile

Shots are imaged via X-ray radiography. The target is irradiated by X-rays produced by a back lighter. In Fig.2, we can see that areas of our shots are undriven i.e. there are no perturbations. This is due to X-ray radiation which has not passed through the target. We are able to produce an image of the X-ray profile by using a low pass filter on the frequency transform of the shot.

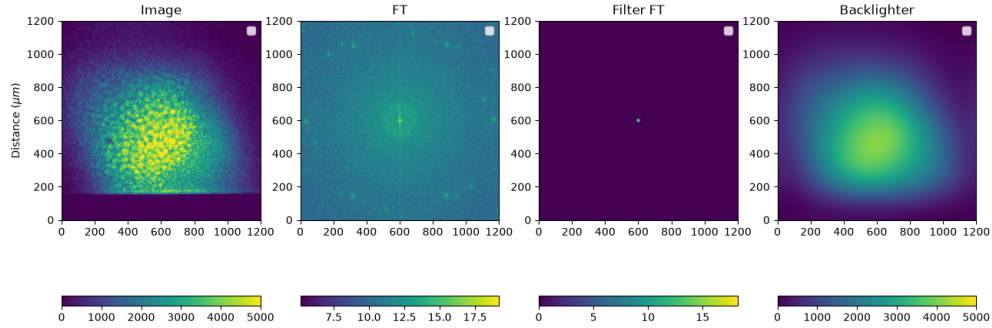


Figure 5: Process of low-pass filtering a raw radiographic image with a Haan window.

Perturbations will contribute to high frequency detail in the Fourier transform. By using a low pass filter, we can filter out this structure to see the back lighter profile. The 2D Fourier transform is computed using the numpy fft library. ‘fft’ is a well-known algorithm for efficiently calculating the Fourier transform of a discrete space. All frequency spectra are centred about the central frequency. A Haan window is utilised as a low pass filter, and the process is shown in Fig.5.

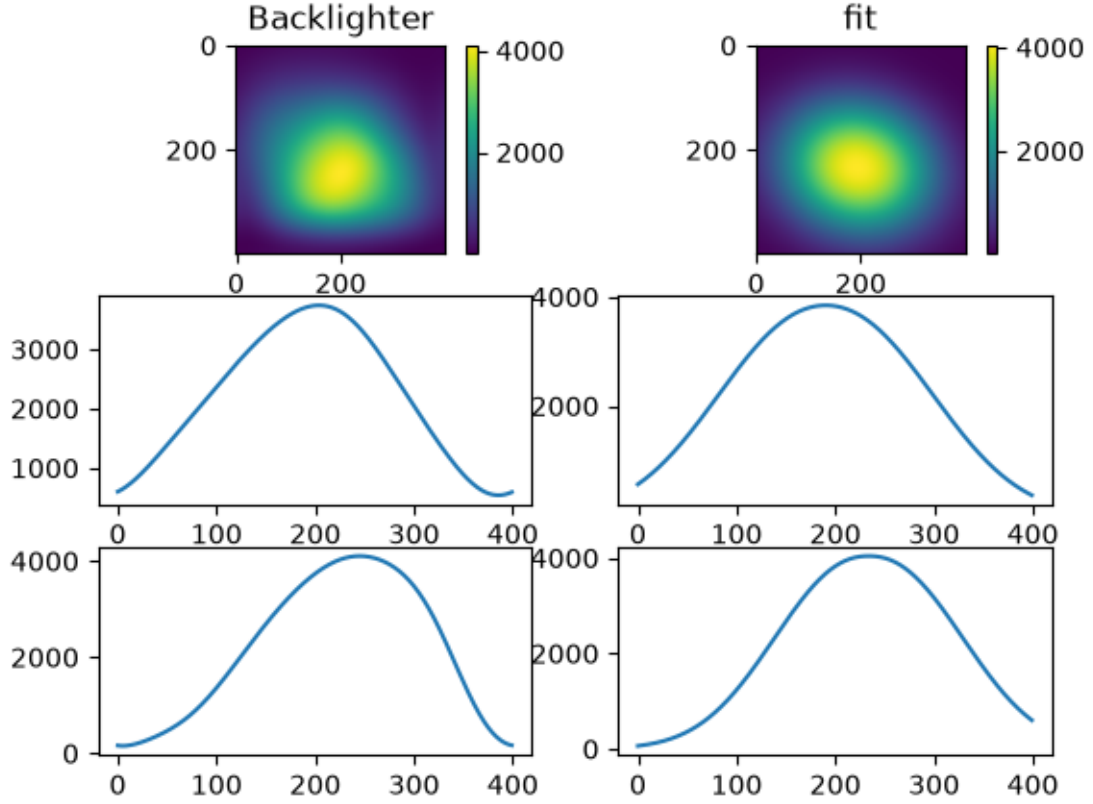


Figure 6: The fit of an elliptical, rotatable super-Gaussian to the back lighter profile of a shot from strip 4, shot 117321.

We can then fit an elliptical, rotatable super-Gaussian to this image. This is done using Scipy optimisation. The result of such a fit can be seen in Fig.6. We complete a fit for each individual shot. A fit is used as the back lighter profile as a low-pass filtered image will still contain some level of structure.

Once we have the back lighter profile of each strip, we can subtract it from each image in order to be left with only the perturbations. Note that this is not the same thing as a transmission image, which we shall discuss in the next section. Fig.7 shows these ‘clean’ shots.

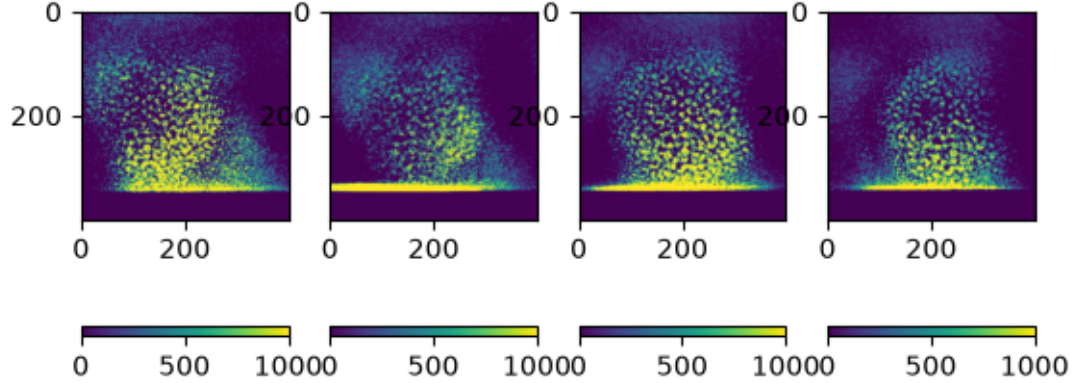


Figure 7: Resultant shot images from strip 4 once the individual back lighter profiles have been removed.

2.3 Correlation

Shots in a strip are produced by a ‘smear technique’ through an array of pinholes. These images are produced within very small timescales (adjacent images have a time step in the order of picoseconds), short enough that hydrodynamic instabilities do not have time to develop. The images are hence comparable, however we have to insure that the details contained within the images overlap.

To do this, we correlate the images. This is simple in theory but is difficult to implement practically. We are trying to match the positions of details across two images. A ‘detail’ is sampled from one image. This is done by cropping the image array around a noticeable detail, for example a bubble. This area is referenced as the region of interest. Then the `signal.correlate2d` function is used to compare the sample image and the adjacent image. This will return an array whose maximal value will represent the area which matches this detail. From there, the difference in position of the detail between the two images can be easily calculated, and then the second image can be reshaped in order to correct this shift. This process can be seen in Fig.8, and Fig.9 displays the final result of the correlation process on a strip.

In practise this is a difficult process. The `signal.correlate2d` function struggles to find a match between images with a large amount of details. Failing to find a match, the function will attempt to find a region which follows the same intensity distribution trend. This causes erroneous matches.

There is no simple way to rectify this issue, one must simply chose a detail which is clear across all images, does not leave the frame due to parallax and iterate sample size until a reasonable match is obtained.

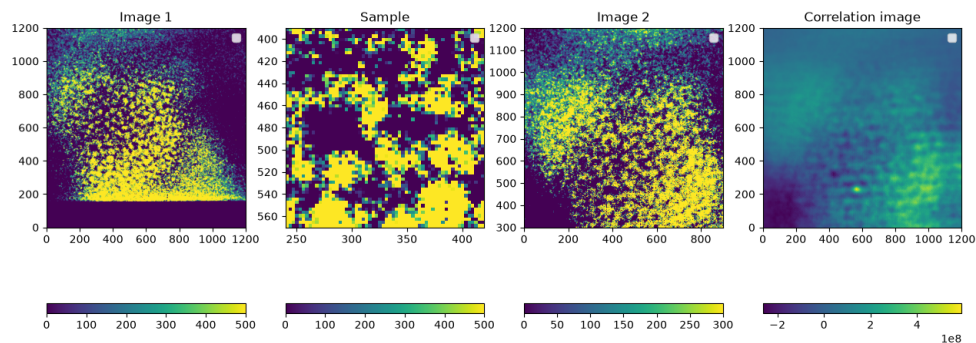


Figure 8: Correlation of images 1 and 2 from strip 4. A sample from image 1 is taken, and the correlated with image 2 in order to find the detail within the image.

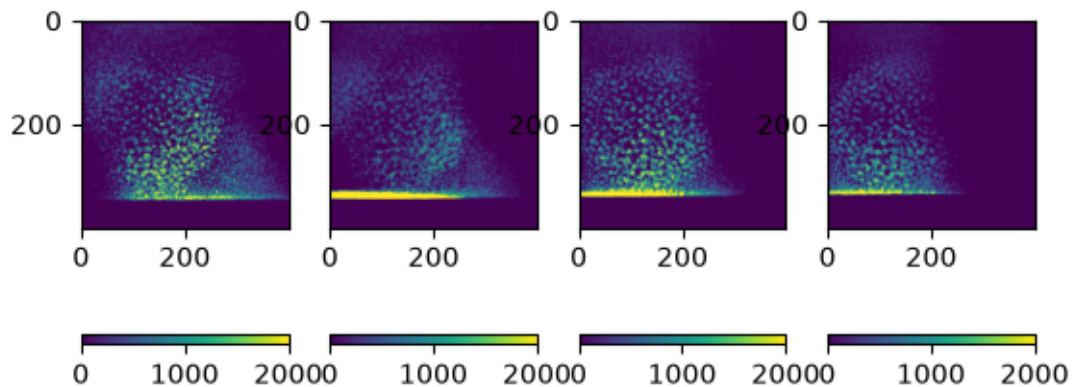


Figure 9: Correlated shots 1,2,3 and 4 from strip 4.

3 Wiener deconvolution

3.1 Theory

The primary source of error in this experiment is from the imaging process. The error can be easily understood if we think in terms of Fourier analysis. The X-rays capture an image which can be described by a 2D function in space, $f(x, y)$. As the rays move towards the CCD camera, the rays will pass through the other pieces of equipment; the pinhole array, the framing camera and the digitisation. Each part has its own unique point spread function, PSF, which can be thought as the optical response to rays which leaves a distinct effect on the image. This is outlined schematically in Fig.10.

In Fourier analysis we can work in either spatial space or frequency space. One can move between the two by using the Fourier transform and the inverse Fourier transform. In spatial space, the image will be given by

$$f'(x, y) = f(x, y) * PSF_{sys}, \quad (1)$$

where PSF_{sys} is the total point spread of the system, given by

$$PSF_{sys} = PSF_{ph} * PSF_{fc} * PSF_{dg}, \quad (2)$$

where PSF_{ph} is the PSF of the pinhole array, PSF_{fc} is the PSF of the framing camera and PSF_{dg} is the PSF of the digitisation.

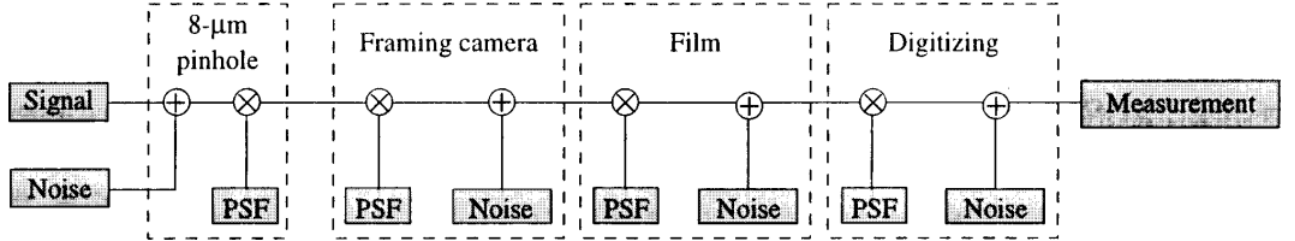


Figure 10: An example of the experimental arrangement with the contributions to the error displayed. This figure is taken from Smalyuk et al [3]

Convolutions are complex to calculate, and in this instance require the computation of a 2D integral. Thankfully, we can switch to frequency space, in which our convolutions become multiplications, utilising the following relation,

$$\mathcal{F}[f(x, y) * g(x, y)] = F(u, v) \cdot G(u, v), \quad (3)$$

where $\mathcal{F}$ indicates the operation of a Fourier transform, $F(u, v), G(u, v)$ represents the Fourier transforms of the spatial functions and u, v are the spatial frequencies in the x and y directions, respectively. Using this formulation, we can transform PSF_{sys} into frequency space. It's Fourier transform, MTF_{sys} , is given by

$$MTF_{SYS} = MTF_{ph} \cdot MTF_{fc} \cdot MTF_{dg}. \quad (4)$$

The formulation of these errors has followed the logic outlined by Smalyuk et al [3]. Smalyuk went further than quantifying these errors, and outlined a methodology to mitigate them via Wiener deconvolution. A simplistic explanation for the method is as follows. Consider an image which is captured by an object with no noise. As discussed above, the image will be effected by the response of the system, quantified by the PSF/MTF. Working in frequency space, we can easily remove the effect of this response by dividing out the MTf, i.e.

$$P(f) = \frac{C(f)}{MTF(f)}, \quad (5)$$

where $P(f)$ is the restored signal and $C(f)$ is the measured signal. Once we factor in noise, this model becomes more nuanced. The measured signal will now be given by $C(f) = S(f) + N(f)$, where $S(f)$ is the signal, and $N(f)$ is the the noise. In order to remove the noise's contribution, we must act a filter on $C(f)$. The filter can be easily derived [ref textbook], and leads to the following equation,

$$P(f) = \frac{C(f)}{MTF(f)} \frac{|S(f)|^2}{|S(f)|^2 + |N(f)|^2}. \quad (6)$$

This equation can be easily implemented. We know turn our focus to experimentally determining each variable.

3.2 Transmission, signal and noise images

The easiest variables to obtain from equation [6] are $C(f), S(f)$ and $N(f)$, using the correlated images, as seen in section 2.3. We firstly add back in the

back lighter profile removed from the image, accounting for the shift imposed by auto-correlation.

From these images we produce transmission images. Transmission, T , is defined as the ratio of the measured intensity over the incident intensity. In our experiment, this is given by our radiograph images over our back lighter profile.

We are now able to produce the signal and noise images. Adjacent transmission images are added and subtracted to produce the corresponding signal and noise transmission images, respectively. We then multiply these images by an average back lighter profile to produce radiographic signal and noise images. We can then take the Fourier transform to produce $S(f)$ and $N(f)$, allowing us to then easily calculate $C(f)$.

3.3 MTF

For our experiment, we have three contributions to the MTF; the pinhole array, the framing camera and the digitisation. For our experiment, the effect of digitisation is minimal and hence can be ignored.

The framing camera MTF has been characterised previously in Smalyuk et al [3], and can be defined as

$$MTF_{fc}(f) = \alpha_1 \exp[-(\sigma_1 \frac{f}{M})^2] + \alpha_2 \exp[-(\sigma_2 \frac{f}{M})^2], \quad (7)$$

where f is the frequency, and $\alpha_1 = 0.89 \pm 0.01$, $\alpha_2 = 0.22 \pm 0.01$, $\sigma_1 = 105.4 \pm 0.4 \mu m$, $\sigma_2 = 2356.8 \pm 0.4 \mu m$ and M is the magnification. The α, σ parameters are found via an experimental fit to the response of the camera to a sharp edge (results from Smalyuk et al [3]).

The pinhole array MTF is created computationally from code provided by Luke Ceurvorst [maybe, passed on by Rob, check]. There is a subtlety to the creation of the MTF. The PSF takes the form of an integral. In order for the integral to converge, it must be evaluated at very high spatial resolution. Once this has been evaluated, one can then interpolate the PSF to lower resolution of our image. One can then simply take the Fourier transform to then acquire the MTF for the pinhole.

3.4 Frequency spectra

A quick aside is needed before we move on to the final piece of image analysis. Our processing has been done within the confines of Fourier analysis. It would be useful to obtain a frequency spectrum of the image. This is straight forward in 1D space, however it becomes more nuanced in 2D space.

A pixel image is discrete, hence we perform a discrete Fourier transformation to get the 2D Fourier transform (as mentioned previously, an efficient algorithm used in many libraries is the fast Fourier transform, FFT []). This will produce a 2D array of the same dimensions, with each entry corresponding to a frequency, or k value. The immediate questions are how do we match the amplitudes to the corresponding k values, and what are the k values.

We can think of k as a vector, with two components; k_x and k_y . Fig.11 illustrates this k space. It becomes immediately apparent that in a discrete 2D space, we have discrete and degenerate k values. We can see that k is the radius of the 2D circle formed, which is constant with respect to the azimuthal angle, ϕ , and varies with k_x and k_y . Hence, we land upon the concept of azimuthal binning; for a certain value of k , we sweep across the k space and bin the different values of k_x and k_y corresponding to that radius. From there, we can then figure out which Fourier amplitudes correspond to which k values (a personal aside-the phrase azimuthal binning has been coined in literature; I think radial binning is much more intuitive, as we are really interested in the radial value and how it varies across the k space).

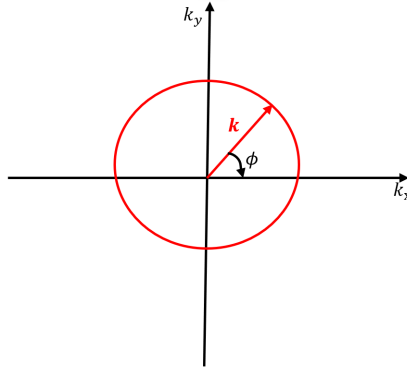


Figure 11: Illustration of the 2D k space.

We now consider the practicalities of azimuthal binning. A 2D array of k values can easily be made provided we have the discrete spatial x and y

axis. From these, we can easily make the corresponding 1D k_x and k_y axis (python has the function `np.fft.ffreq`). To account for degeneracy, we pick one k_x value, add in quadrature with each k_y value, and then repeat for each k_x value. This will produce a 2D array of all our k values. Borrowing the notation of matrices, the F_{ij} entry of the Fourier transform matrix F will be the corresponding Fourier amplitude of the K_{ij} entry of the k value matrix K just formed. One can then sort the arrays accordingly to account for degenerate k values.

3.5 Deconvolution

We now have all the components required to produce a deconvolved image. There is some nuance however when we consider the signal and noise components. If we plot the frequency spectra of both on the same axis, we can see that the two images overlap after a certain value of k . This is shown in Fig.12. Physically, this means we cannot distinguish between the signal and the noise, and hence we apply a low-pass filter to the FT of the deconvolved image. The reconstructed image is shown in Fig.13

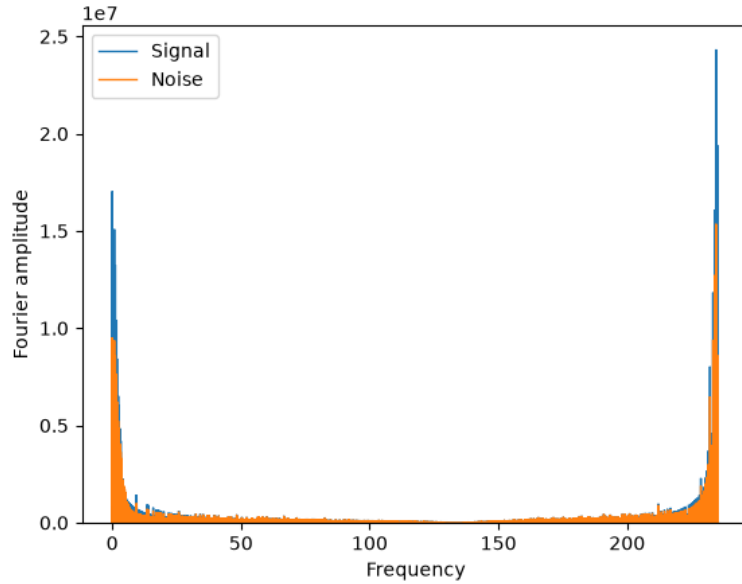


Figure 12: Frequency spectra of the FT of the signal and noise images.

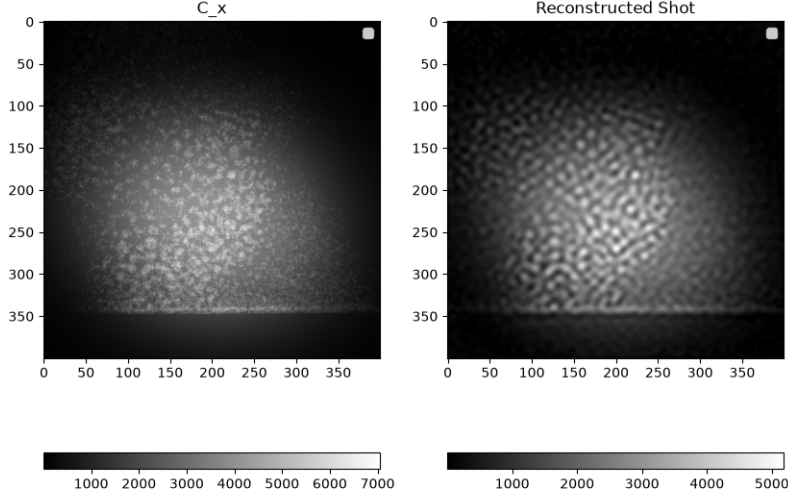


Figure 13: Comparison of the raw measured image with the Wiener deconvolved image. The measured image is formed from images one and two from strip 4.

4 Optical Density

A radiograph measures the the transmitted intensity of X-ray radiation through the target. The ratio of the incident intensity, I_0 , to the measured intensity, I , and be modelled by an exponential decay,

$$\frac{I}{I_0} = e^{-\tau}, \quad (8)$$

where τ is the optical depth. In literature, it is not uncommon to refer to the ratio as the transmission.

The optical depth is a useful parameter as it can be linked to physical quantities. The transmission can also be modelled by considering the physical

path the rays travelled through in the target,

$$\frac{I}{I_0} = e^{-\mu\rho d}, \quad (9)$$

where μ is the attenuation of the beam, ρ is the compressed density of the target, and d is the distance travelled through the target. Equating the powers of the exponentials, we get the following relation,

$$\tau = \mu\rho d. \quad (10)$$

With the above relation for optical density, we can use our deconvolved images to obtain a physical measurement of the perturbation size distribution in the target. By dividing the image by the average back lighter profile, and then taking the negative of the logarithm of this ratio gives the OD image, as seen in Fig.14. The rays must travel through the perturbations, or 'bubbles', which have a height, S_k . Hence, by replacing d with S_k in the above equation, and by taking FT of the OD image and using azimuthal binning, we get a distribution of the perturbation size, as seen in Fig.15 below.

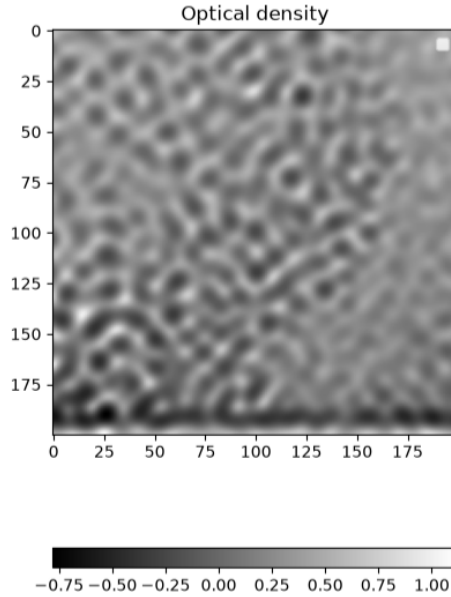


Figure 14: Optical density image produced from the Weiner deconvolved image produced from images 1 and 2 from strip 4.

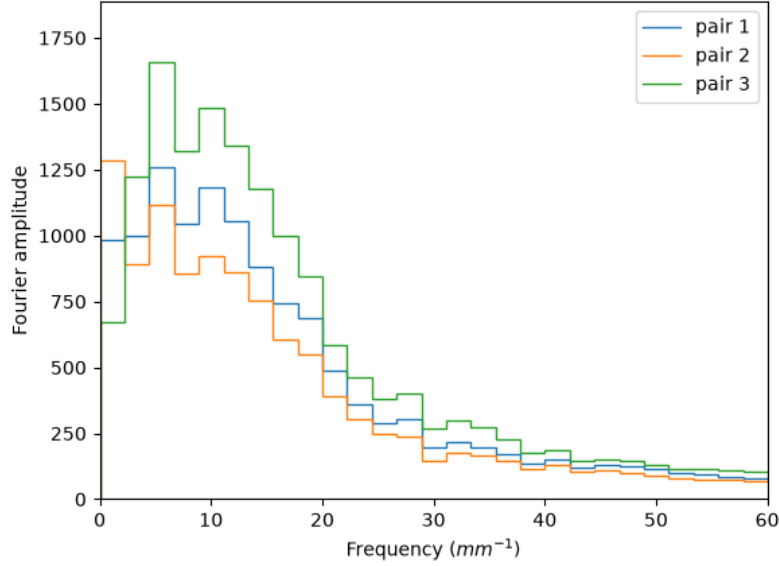


Figure 15: Frequency spectrum of the optical density image produced from the Weiner deconvolved image produced from images 1 and 2 from strip 4.

References

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